CMUC Online Seminar on Algebra, Logic and Topology

University of Coimbra

Abstract and concrete spectral spaces

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Abstract

In this joint work with Alberto Facchini and Carmelo Antonio Finocchiaro we begin with a complete lattice L equipped with a binary multiplication such that the product of any two elements is less or equal to their meet. As suggested by the so-called "Abstract ideal theory" we define the spectrum Spec(L) as the space of prime elements in L and prove that Spec(L) is a sober space. Then we discuss sufficient conditions for Spec(L) to be a spectral space. We consider various examples, and in particular propose a new kind of 'non-commutative topology' based on commutator theory.